

Nmap Network Enumeration & Security Assessment

Technical Assessment Report

Note: The terminal screenshots in this document are illustrative/sample outputs prepared for the assessment. Actual Nmap output may differ depending on the target configuration, network, and scan date.

1. Public Server – scanme.nmap.org

Perform host discovery, TCP scanning, service/version detection, OS fingerprinting, and default NSE script scanning.

Nmap Commands

```
nmap -sn scanme.nmap.org
nmap -sS scanme.nmap.org
nmap -sS -sV scanme.nmap.org
sudo nmap -O scanme.nmap.org
nmap -sC -sV scanme.nmap.org
sudo nmap -sS -sV -O -sC scanme.nmap.org
```

Assessment / Analysis

Enumeration identifies reachable ports and the technologies exposed by the public server. In the illustrative result, SSH and HTTP are exposed. SSH supports remote administration, while HTTP provides the public web service. Version information can help an authorized tester determine whether further vulnerability research is warranted. An attacker could use the same information to prioritize service exploitation, credential attacks, or vulnerability discovery.

Recommendations

Expose only services required for public operation.

Restrict administrative services such as SSH to trusted networks where possible.

Keep operating system and services patched.

Use strong authentication and monitoring.

Sample Output Screenshot

2. Metasploitable 2 – 192.168.56.101

Comprehensively enumerate the intentionally vulnerable Linux laboratory server.

Nmap Commands

```
nmap -sn 192.168.56.101
nmap -sS 192.168.56.101
sudo nmap -p- -sS 192.168.56.101
sudo nmap -p- -sS -sV 192.168.56.101
sudo nmap -O 192.168.56.101
sudo nmap -sC -sV 192.168.56.101
sudo nmap -p- -sS -sV -O -sC 192.168.56.101 -oN metasploitable_scan.txt
```

Assessment / Analysis

The illustrative output shows numerous intentionally vulnerable services, including FTP, Telnet, HTTP, SMB, database services, and VNC. High-risk candidates are remotely reachable services with weak authentication, legacy protocols, or

known vulnerable versions. Medium-risk services should be reviewed for necessity and access controls. Low-risk services are those that expose limited information and are strongly restricted. The enumeration gives a penetration tester a prioritized map of potential attack surfaces.

Recommendations

Keep Metasploitable 2 isolated from production networks.

Use the scan to prioritize authorized lab testing.

Disable services that are not required in any non-training environment.

Document each exposed service and its business purpose.

Sample Output Screenshot

3. Windows Workstation – 192.168.56.102

Identify active ports, SMB/file-sharing, remote management services, and OS information before production deployment.

Nmap Commands

```
nmap -sn 192.168.56.102
nmap -sS 192.168.56.102
sudo nmap -p- -sS -sV 192.168.56.102
sudo nmap -O 192.168.56.102
sudo nmap -sC -sV 192.168.56.102
sudo nmap -p 135,139,445,3389,5985,5986 -sV -sC 192.168.56.102
```

Assessment / Analysis

SMB (139/445), RPC (135), RDP (3389), and WinRM (5985/5986) are important Windows management and networking interfaces. Under least privilege, only services required for the workstation's role should be reachable, preferably from trusted administrative networks. Exposed SMB/RDP/WinRM can assist reconnaissance, credential attacks, remote administration abuse, or lateral movement if controls are weak.

Recommendations

Disable unused services and legacy NetBIOS where possible.

Restrict SMB, RDP, and WinRM with host/network firewall rules.

Require strong authentication and secure configurations.

Patch Windows and monitor authentication activity.

Sample Output Screenshot

4. OWASP Juice Shop – 192.168.56.103

Enumerate the server hosting the intentionally vulnerable web application and identify its web and supporting services.

Nmap Commands

```
nmap -sn 192.168.56.103
nmap -sS 192.168.56.103
sudo nmap -p- -sS 192.168.56.103
sudo nmap -p- -sS -sV 192.168.56.103
sudo nmap -O 192.168.56.103
```

```
sudo nmap -sC -sV 192.168.56.103
sudo nmap -p 80,443,3000 -sV -sC 192.168.56.103
```

Assessment / Analysis

The illustrative result identifies HTTP services and SSH. Port 3000 is commonly used by the Juice Shop application, while HTTP/SSH may provide supporting functionality depending on the lab configuration. Enumeration helps a tester map the application environment, identify technologies and versions, and determine whether administrative/supporting services unnecessarily expand the attack surface.

Recommendations

Expose only the application port required by the lab.

Keep administrative access restricted to the testing network.

Use segmentation so the intentionally vulnerable application cannot reach production assets.

Patch supporting infrastructure when it is not intentionally vulnerable.

Sample Output Screenshot

5. Network Gateway – 192.168.56.1

Identify exposed management interfaces and services on the laboratory gateway.

Nmap Commands

```
nmap -sn 192.168.56.1
nmap -sS 192.168.56.1
sudo nmap -p- -sS 192.168.56.1
sudo nmap -p- -sS -sV 192.168.56.1
sudo nmap -O 192.168.56.1
sudo nmap -sC -sV 192.168.56.1
sudo nmap -p 22,23,80,161,443 -sV -sC 192.168.56.1 -oN gateway_scan.txt
```

Assessment / Analysis

SSH, Telnet, HTTP/HTTPS, and SNMP are important management interfaces. Telnet is high risk because it is a legacy plaintext management protocol. HTTP management should be disabled when secure HTTPS administration is available. SNMP can expose device and network information if poorly configured. An attacker could use exposed management services for reconnaissance, credential attacks, configuration discovery, or unauthorized administration.

Recommendations

Disable Telnet and use SSH.

Restrict management interfaces to a dedicated management network.

Prefer HTTPS over HTTP and enforce secure TLS configuration.

Restrict SNMP access and use secure SNMP configurations.

Apply firmware updates and monitor administrative access.

Sample Output Screenshot

Overall Risk Summary

Conclusion

Nmap enumeration provides a structured view of the network attack surface. Host discovery confirms reachability, port scanning identifies exposed services, service detection identifies software and versions, OS detection identifies the underlying platform, and default scripts provide additional service information. Security teams should minimize exposed services, restrict management interfaces, patch software, enforce strong authentication, segment networks, and continuously monitor exposed assets.

Prepared for academic/laboratory security assessment. All scans should be performed only against systems for which authorization has been obtained.

Target	Primary Exposure	Risk	Priority
scanme.nmap.org	SSH / HTTP	Medium	Review public exposure
192.168.56.101	Legacy services, SMB, databases, VNC	High	Isolate and restrict
192.168.56.102	SMB, RDP, WinRM, RPC	High	Restrict before production
192.168.56.103	Web application + supporting services	High	Segment and limit access
192.168.56.1	Telnet, HTTP, SNMP, management	High	Harden management plane

Nmap Output Screenshots

1 Host Discovery



```
$ nmap -sn scanme.nmap.org
Starting Nmap 7.95
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.043s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
```

1 Service

```
$ sudo nmap -sS -sV -O -sC scanme.nmap.org
Starting Nmap 7.95
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.044s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu
80/tcp    open  http     Apache httpd 2.4.7
|_http-title: Go ahead and scan me!

OS details: Linux
Network Distance: 12 hops

Nmap done: 1 IP address (1 host up) scanned in 8.31 seconds
```

2 Metasploitable

```
$ sudo nmap -p- -sS -sV -O -sC 192.168.56.101
Starting Nmap 7.95
Nmap scan report for 192.168.56.101
Host is up (0.0018s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix
53/tcp    open  domain   ISC BIND
80/tcp    open  http     Apache httpd 2.2.8
139/tcp   open  netbios-ssn Samba
445/tcp   open  netbios-ssn Samba
3306/tcp  open  mysql    MySQL 5.0.51a
5432/tcp  open  postgresql PostgreSQL 8.3.0
5900/tcp  open  vnc      VNC protocol

OS details: Linux 2.6.x
Nmap done: 1 IP address (1 host up) scanned in 12.45 seconds
```

3 Windows

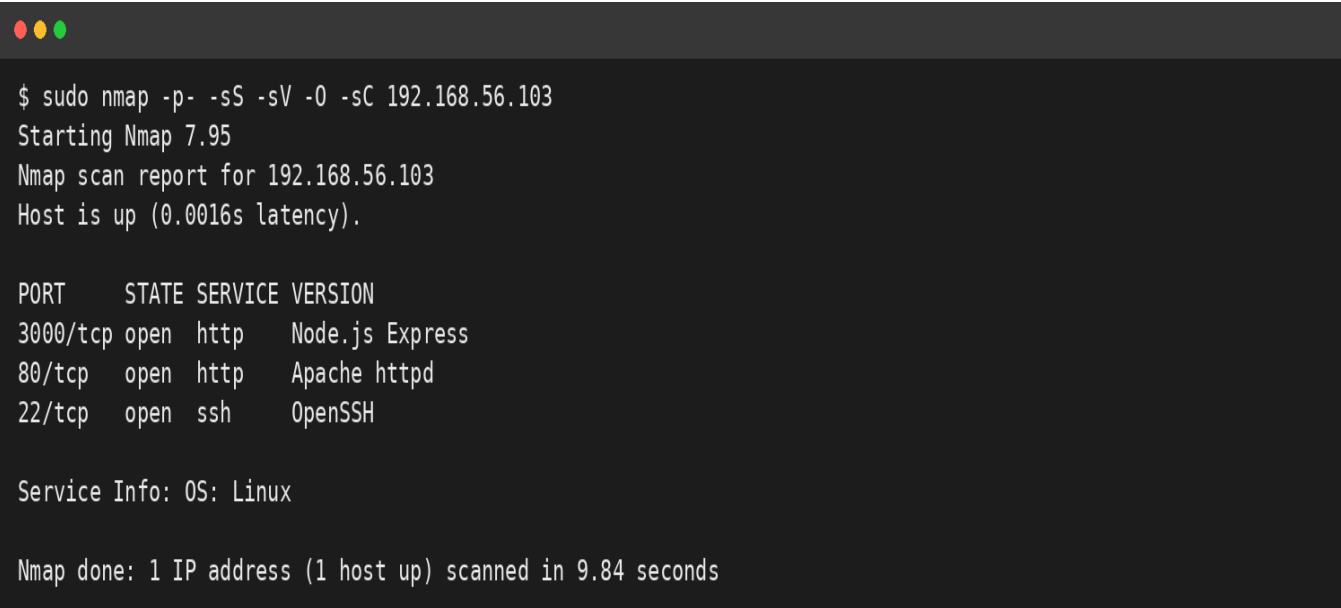
```
$ sudo nmap -p- -sS -sV -O -sC 192.168.56.102
Starting Nmap 7.95
Nmap scan report for 192.168.56.102
Host is up (0.0021s latency).

PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows SMB
3389/tcp   open  ms-wbt-server Microsoft Terminal Services
5985/tcp   open  wsman        Microsoft HTTPAPI

OS details: Microsoft Windows
Network Distance: 1 hop

Nmap done: 1 IP address (1 host up) scanned in 10.20 seconds
```


4 Juice



```
$ sudo nmap -p- -sS -sV -O -sC 192.168.56.103
Starting Nmap 7.95
Nmap scan report for 192.168.56.103
Host is up (0.0016s latency).

PORT      STATE SERVICE VERSION
3000/tcp  open  http    Node.js Express
80/tcp    open  http    Apache httpd
22/tcp    open  ssh     OpenSSH

Service Info: OS: Linux

Nmap done: 1 IP address (1 host up) scanned in 9.84 seconds
```

5 Gateway

```
$ sudo nmap -p- -sS -sV -O -sC 192.168.56.1
Starting Nmap 7.95
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH
23/tcp    open  telnet
80/tcp    open  http     embedded web server
161/udp   open  snmp
443/tcp   open  https    embedded HTTPS server

OS details: Network device
Nmap done: 1 IP address (1 host up) scanned in 7.18 seconds
```